Machine Learning Project Guidelines

General Rules

- Group size-5. Only one group member should submit the assignment and must tag the rest of the team members.
- If teammates are not tagged or multiple submissions are made, you may submit a regrade request, but a **1-point deduction** will be applied.
- All submissions are to be made via **Google Classroom**.

Project Types

You may choose among the following categories:

1. **Application Project**: Apply machine learning algorithms to a real-world problem or dataset.
2. **Algorithmic Project**: Develop a new learning algorithm or extend an existing one.
3. **Theoretical Project**: Prove interesting or non-trivial properties of ML algorithms.

Projects may also combine application, algorithmic, and theoretical aspects.

Resources

- Major ML conferences for project ideas:
  - ICML (2022–2026): `https://icml.cc`
  - NeurIPS (2021–2025): `https://nips.cc`
  - ICLR (2022–2026): `https://iclr.cc/`
  - CVPR (2022–2026): `https://openaccess.thecvf.com/menu`
- Academic search engine: `http://scholar.google.com`
- Datasets: Kaggle, UCI ML Repository, or self-collected datasets (note preprocessing effort).

Project Components and Deadlines

1. Proposal (Deadline: 30 September)

- Format: PDF, 300–500 words
- Content:
  - Title and project category
  - Full names and enrollment numbers of all team members
  - **Motivation:** What problem are you tackling and why?
  - **Method:** What ML techniques will you apply or develop?
  - **Intended Experiments:** Baselines, evaluation strategy, expected results
  - **Contributions:** Describe each member's role

2. Milestone (Deadline: 20 October)

- Format: At most 3 pages (excluding references)
- Content:
  - Motivation (brief restatement)
  - Methods tried so far and justification
  - Preliminary experiments, baselines, results, error analysis
  - Next steps and planned improvements

3. Poster (Deadline: 15 November)

- Format: Will be shared
- Content: Clear and concise visual explanation of problem, methods, and results

4. Final Report (Deadline: 15 November)

- Format: 4–6 pages using ICML/NeurIPS LaTeX style
- Suggested structure:
  1. Abstract
  2. Introduction & Motivation
  3. Related Work
  4. Methodology
  5. Experiments & Results
  6. Analysis & Discussion
  7. Conclusion
  8. References
  9. Contributions (per team member)

Evaluation Criteria

Projects will be graded on:

1. **Technical Quality:** Sound methodology, appropriate algorithms, meaningful results.
2. **Originality:** Novel data processing, analysis, or extensions beyond existing work.
3. **Communication:** Clear explanation of context, methods, results in proposal, milestone, poster, and report.

Notes

- Deep learning projects should also include classical baselines (e.g., logistic regression, SVM, unsupervised methods).
- Using preprocessed datasets is allowed, but data exploration and error analysis are strongly encouraged.
- Replication of prior work is allowed, but must be extended with new applications or analysis.